

Avi-Defender

S M A R T B I R D M O N I T O R I N G &
D E T E R R E N T S Y S T E M

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Skit Time!



💡 "Bird intrusions are a persistent issue, but what if we could stop them intelligently and automatically?"



Skit Time!



🎭 Setup:

- Imagine you're returning a tray at a hawker center.
- You see birds swooping in, scavenging leftover food.
- Messy, unhygienic, and unappealing for diners!



💡 "Bird intrusions are a persistent issue, but what if we could stop them intelligently and automatically?"



The Problem - Why Birds Are a Concern?



Public Health Concerns

- Salmonella Bacteria
- Droppings
- Invasive
- Noisy
- Stomach Pain
- Fever Diseases
- Diarrhoea
- Ornithosis
- Disturbing

Problem? 🗑️ Birds scavenge for food → Contamination risk!

Ref: [CNA Explains: Singapore has a pigeon problem. What's being done about it?](#)



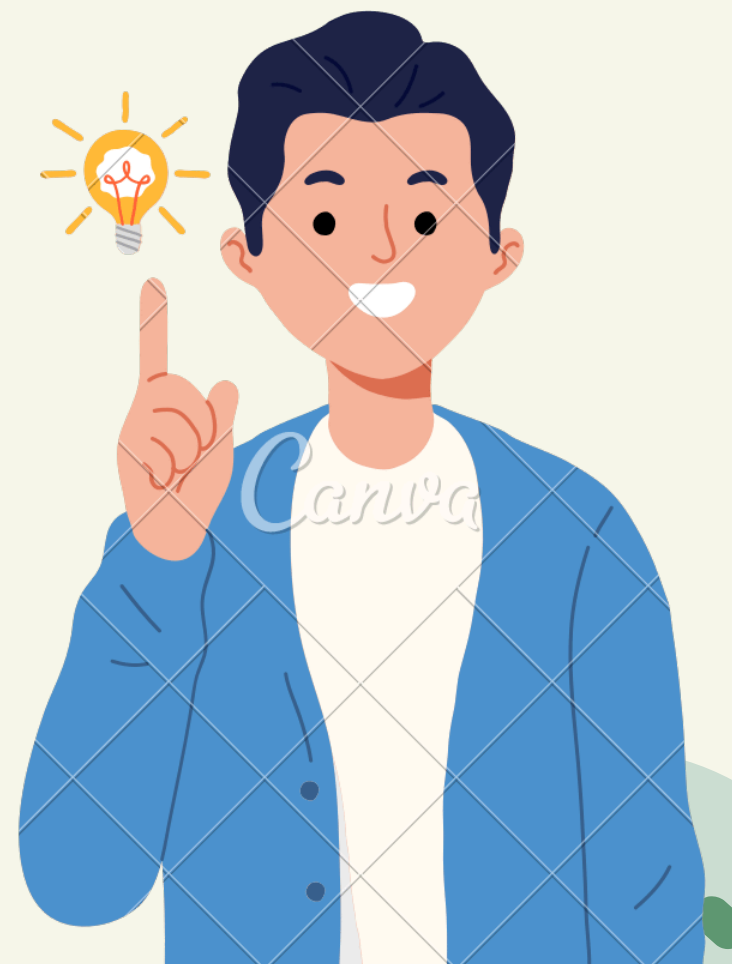
INTRODUCING *Avi-Defender*

 An **IoT-based Bird Detection & Deterrent System** for maintaining hygiene in public spaces.

 **Automates** bird detection and deterrence using **ultrasonic** sensors and **cameras**.

 Provides **real-time** alerts to help staff manage bird activity efficiently.

 **AI-powered** detection & deterrence  **Scalable** for multiple locations



POTENTIAL LOCATIONS, USERS & BENEFICIARIES



Location: Hawker Centres &
Food Courts
Users/Beneficiaries: Stall
owners/ Diners



Location: Outdoor
Restaurants & Theme Parks
Users/Beneficiaries:
Stallowners/Diners/Visitors



Location: Farms
Users/Beneficiaries:
Farm Owners

-  Hands-Free Bird Control – Automates deterrence, freeing up staff.
-  Less Cleanup, More Hygiene – Prevents mess, keeps spaces cleaner.
-  Food-Safe Environments – Stops birds from contaminating open food areas.

PROJECT GOALS/IMPACTS

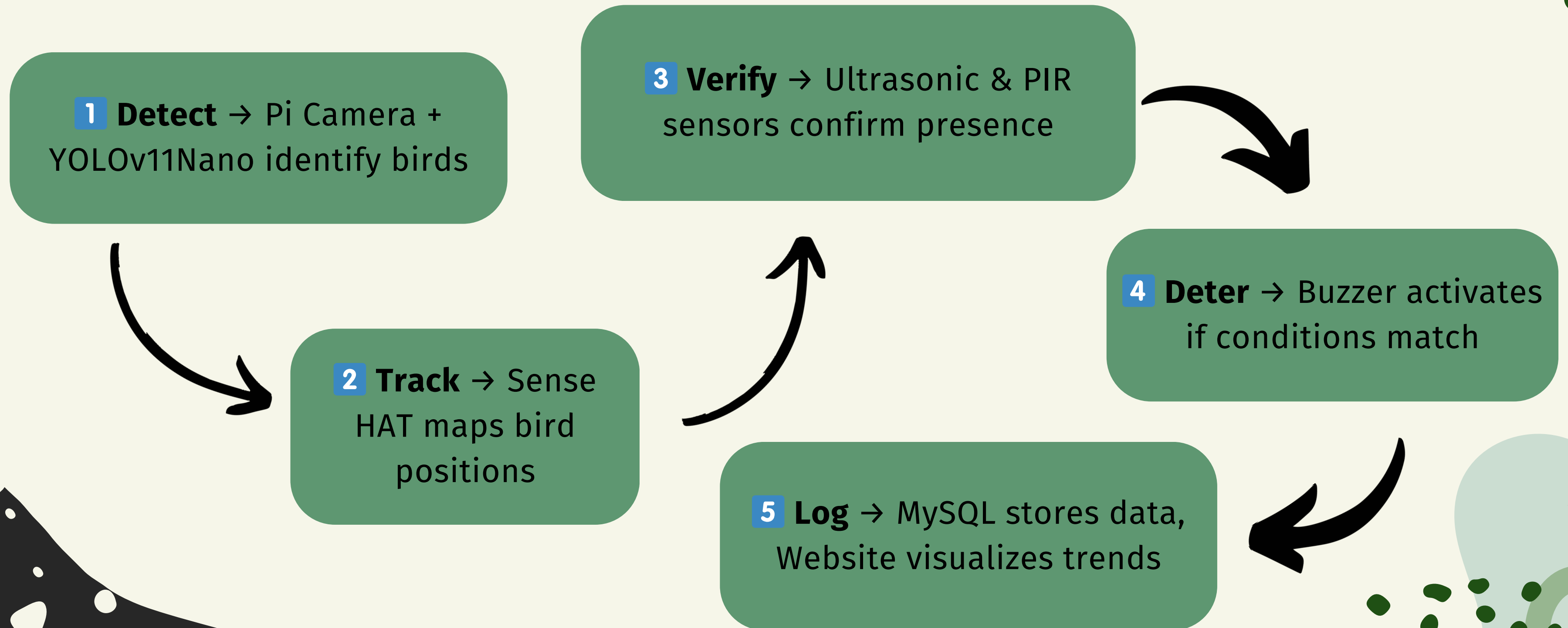
- **Non-harmful** IoT system 🌿
- Bird detection via **ultrasonic sensor & camera** 📡
- **Audio-based** deterrence 🔊
- **Real-time** configuration
- Data logging & insights 💻



- 🏠 **Hygienic** outdoor spaces
- 🌽 **Fewer** crops loss
- 👤 **Efficient** operations for staff



AVI-DEFENDER - END-TO-END PROCESS



RASPBERRY PI FUNCTIONS

🎯 Enhances detection & system management

⚙️ Key Functions:

- 📷 AI bird detection (Camera + YOLOv11n)
- 📊 Logs bird count & location (MySQL)
- 🔄 MQTT communication with Arduino

🔧 Optimisations:

- ✅ Multi-threading for speed
- ✅ TensorFlow Lite for efficiency
- ✅ Simplified live feed (dots & circles)

Raspberry Pi 4

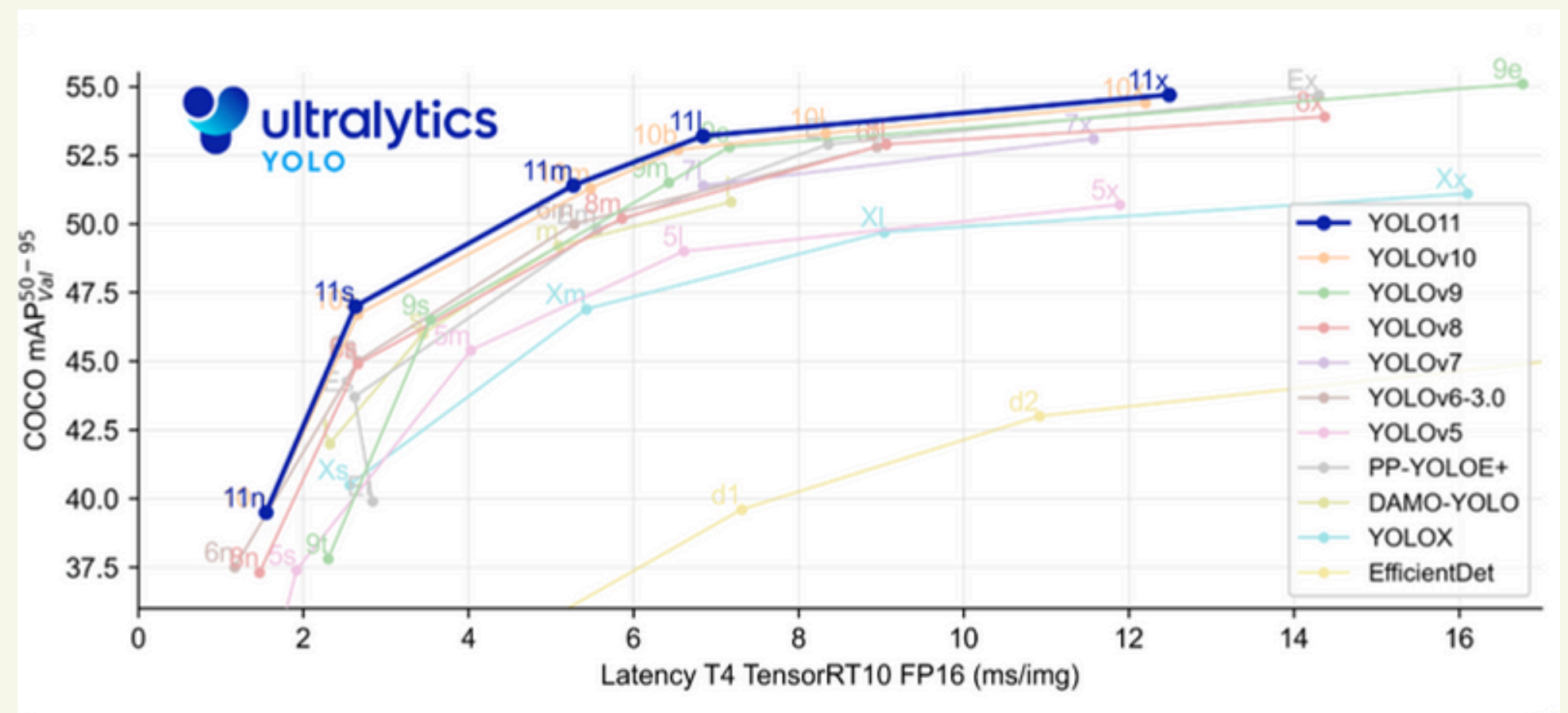


Raspberry Pi
Camera 2

Our Model : YOLOv11Nano

Why Yolov11Nano?

- Lightweight Model Optimized for Edge Devices
- Faster Inference with Lower Computational Cost
- 20 FPS Capable, Limited by Single-Frame Capture
- Trained on 10,000+ Images for Bird Detection





YOLO OPTIMIZATION

Single-thread, tf-lite

The screenshot displays a Raspberry Pi desktop environment. On the left, a terminal window shows system statistics and a process list. The top of the terminal displays: `0[|||||] 4.5% Tasks: 93, 234 thr, 135 kthr; 2 runnin`, `1[|||||] 5.8% Load average: 1.06 1.30 1.15`, `2[|||||] 12.0% Uptime: 00:48:14`, `Mem[|||||] 1.04G/3.70G`, and `Swp[|||||] 15.9M/512M`. Below this is a table of processes with columns: PID, USER, PRI, NI, VIRT, RES, SHR, S, CPU. The bottom of the terminal shows a traceback for a `ValueError: Cannot set tensor: Dimension mismatch. Got 416 but expected 640 for dimension 1 of input 0.` and the command `(iot) myrios@minpi:~/iot/codes $ python yolo_video_single.py`.

In the center, a video feed window titled 'YOLO TFLite Bird Detection' shows a live camera feed of a paved area. Four birds are detected and labeled with blue boxes and text: 'Bird: 0.58', 'Bird: 0.58', 'Bird: 0.58', and 'Bird: 0.58'. The FPS is displayed as 'FPS: 1.17'. At the bottom of the video feed, a bounding box is shown with coordinates: `(x=829, y=88) - R:230 G:239 B:229`.

On the right, a web browser window titled 'Bird Detection Dashboard - Chromium' shows a website with a navigation bar (Home, Project Purpose, Dashboard, Contact) and a main content area with the heading 'Bird Detection and System'. Below the heading is a paragraph: 'Birds Through Innovation'. At the bottom of the browser window, there is a 'Project Purpose' section with the text: 'This IoT project is designed to protect tray return areas by detecting and deterring birds. An Arduino'.



YOLO OPTIMIZATION

Multi-thread, tf-lite

The screenshot displays a multi-window desktop environment. On the left, a terminal window shows system statistics and a process list. The top-right window is a web browser displaying the 'Bird Detection Dashboard'. The bottom-left window shows a code editor with Python code and error messages. The bottom-right window shows a video feed with bird detection overlays.

Terminal Window (Left):

```
0[|||||92.3%] Tasks: 93, 236 thr, 138 kthr; 4 runnin
1[|||||182.7%] Load average: 3.87 1.67 1.17
2[|||||96.6%] Uptime: 00:44:01
3[|||||71.7%]
Mem[|||||2.536/3.700]
Swap[|||||16.0M/512M]

MAIN: 0.00

PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
7761 myrios 20 0 3612M 2182M 318M R 243.7 57.5 3:02.97 python yolo_v
7776 myrios 20 0 3612M 2182M 318M R 79.6 57.5 0:56.70
7794 myrios 20 0 3612M 2182M 318M R 77.9 57.5 0:55.83
810 vnc 20 0 830M 118M 92048 S 44.6 3.1 3:49.63
931 myrios 20 0 587M 182M 60128 S 13.1 2.7 1:29.45
7781 myrios 20 0 308M 82212 57396 R 13.1 2.1 0:10.30
1107 vnc 20 0 830M 118M 92048 S 8.9 3.1 0:30.13
1108 vnc 20 0 830M 118M 92048 S 8.9 3.1 0:31.01
1110 vnc 20 0 830M 118M 92048 S 8.3 3.1 0:30.41
```

Web Browser (Top Right):

Bird Detection Dashboard - Chromium

192.168.125.247:5000

Bird Detection

Home Project Purpose Dashboard Contact

Video Feed (Bottom Right):

YOLO TFLite Bird Detection

FPS: 24.82

Bird 0.88

Bird 0.83

Bird 0.43

(x=106, y=140) ~ 8.179 0.129 8.95

Code Editor (Bottom Left):

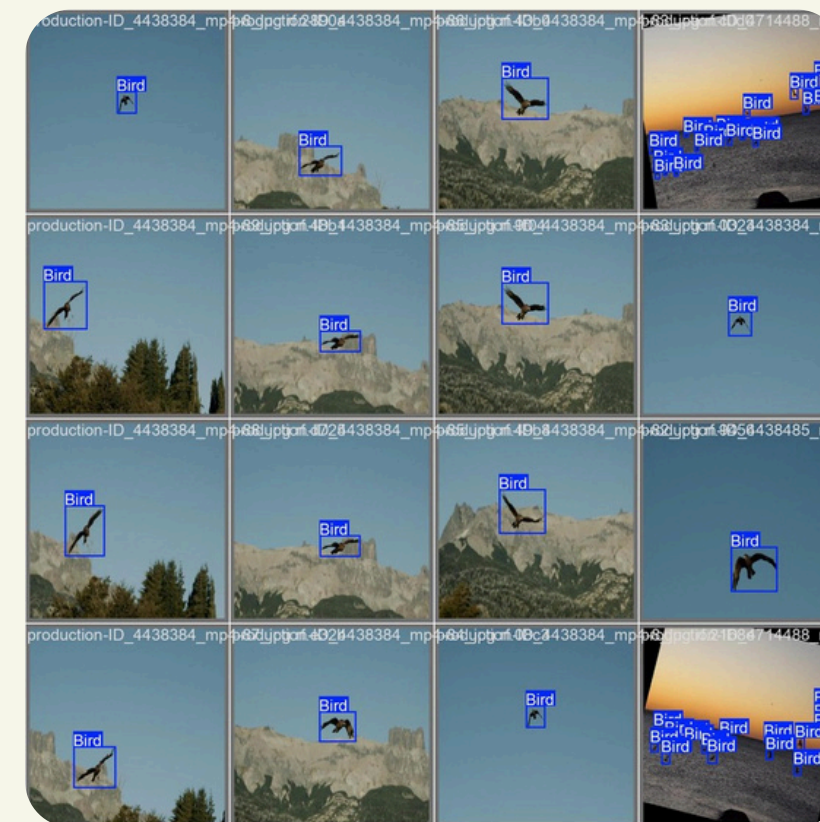
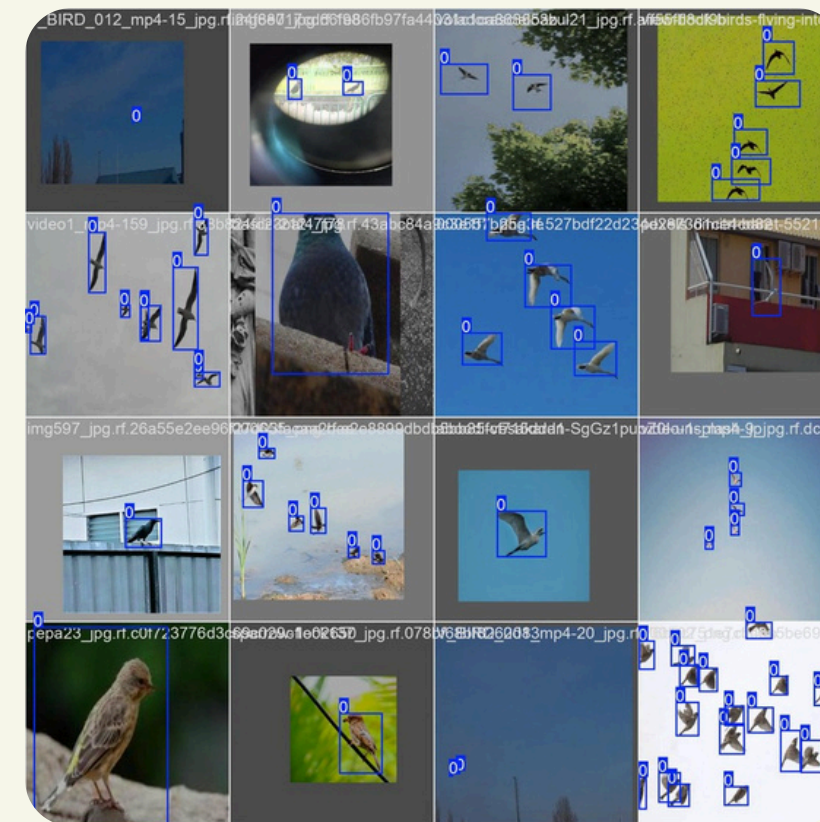
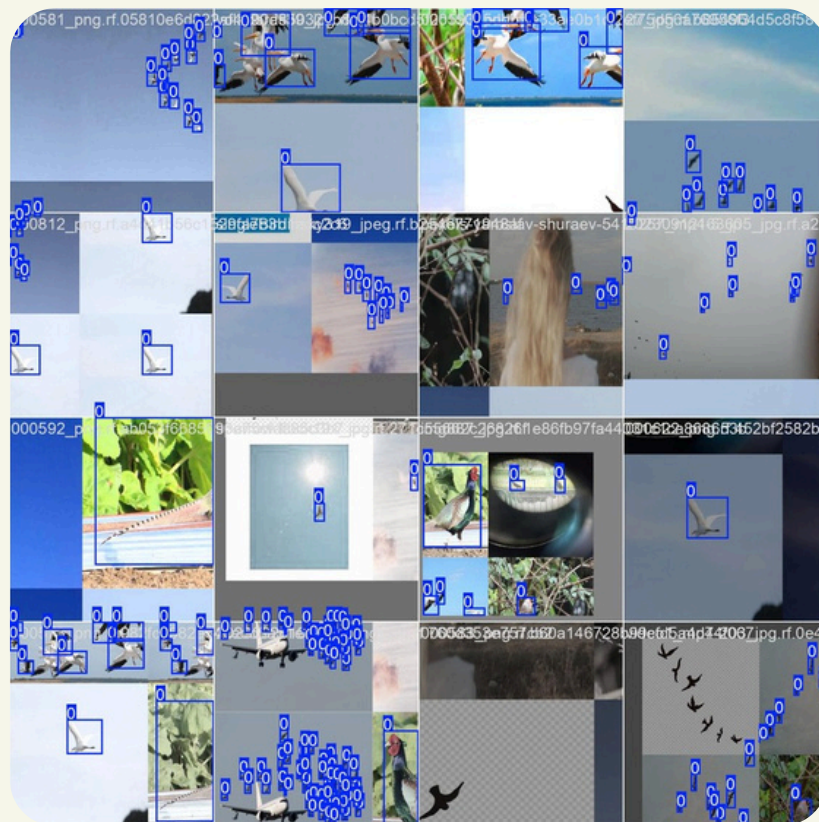
```
erpreter.py", line 73, in __init__
self._library = ctypes.pydll.loadlibrary(library)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
File "/usr/lib/python3.11/ctypes/_init_.py", line 454, in Load
return self._dilltype(name)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
File "/usr/lib/python3.11/ctypes/_init_.py", line 376, in __in
self._handle = _dlopen(self._name, mode)
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
OSError: libtensorflowlite.so: cannot open shared object file: No
irectory
Exception ignored in: <function Delegate.__del__ at 0x7f786ef600>
Traceback (most recent call last):
File "/home/myrios/iot/lib/python3.11/site-packages/tensorflow/L
erpreter.py", line 109, in __del__
if self._library is not None:
AAAAAAAAAAAAAAAA
AttributeError: 'Delegate' object has no attribute '_library'
(iot) myrios@minpi:~/iot/codes $ python yolo_video_multi.py
Error in cpuinfo: precli(PR_SVE_GET_VL) failed
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
qt.qpa.plugin: could not find the qt platform plugin "wayland" in "/home/myrios/
iot/lib/python3.11/site-packages/cv2/qt/plugins"
```

Project Purpose (Bottom Right):

This IoT project is designed to protect tray return areas by detecting and deterring birds. An Arduino

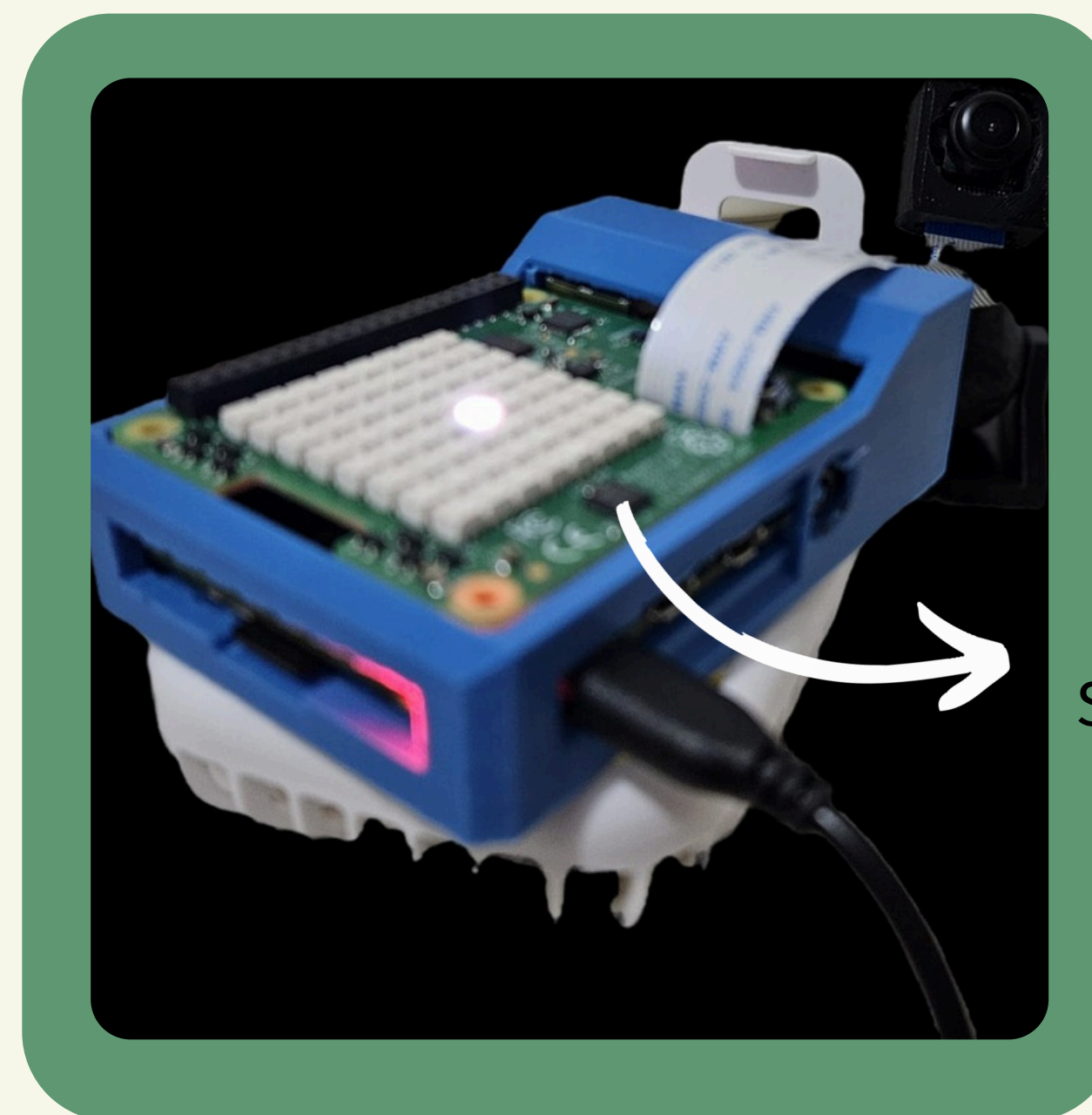


Output Examples



SENSE HAT & REAL-TIME VISUALIZATION

- Maps Detections → Displays bird locations on an 8x8 LED matrix.
- Dynamic Tracking → Blue pixels show presence, intensity indicates proximity.
- 🖼️ Suggested Images:
 - Diagram showing Raspberry Pi's role in detection & data flow.
 - Sense HAT display mapping bird positions

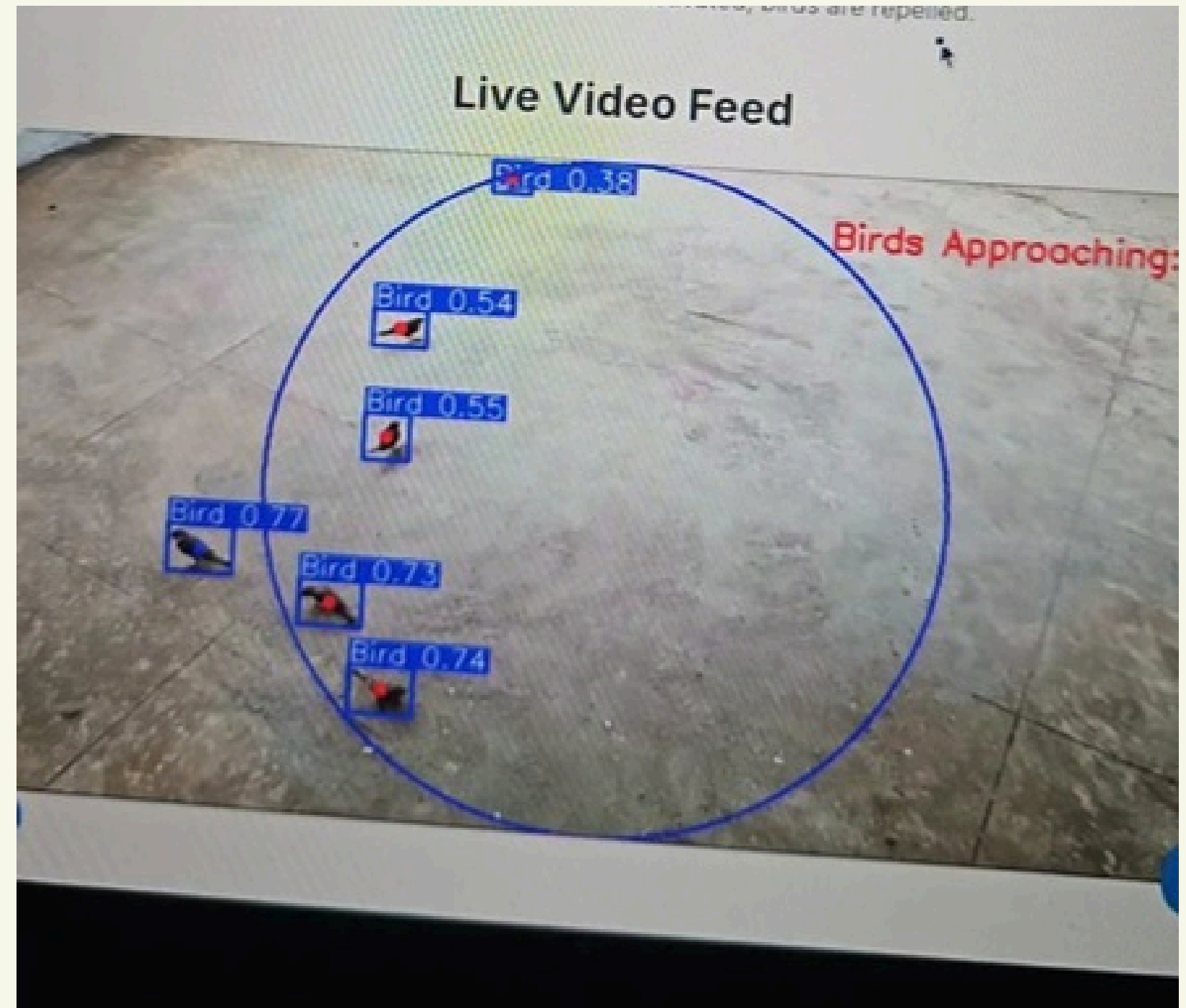


Sensehat



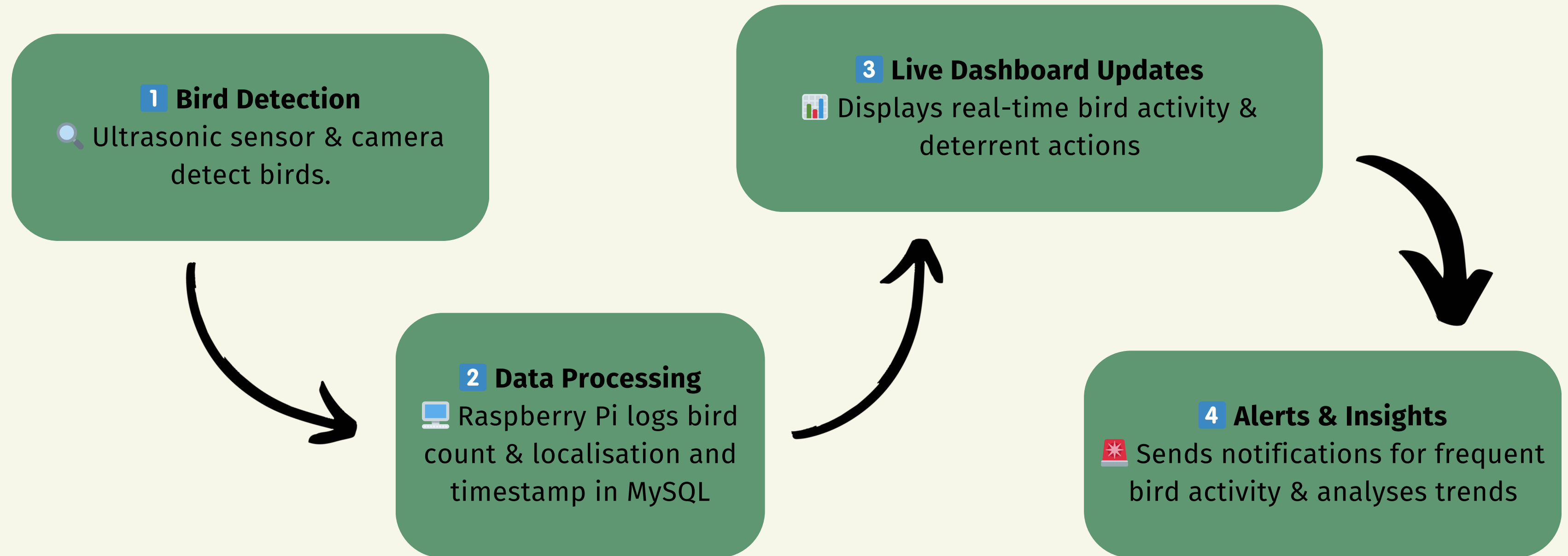
ORIGINAL YOLO

with SenseHat update





REAL-TIME DATA INSIGHTS DASHBOARD FLOW



⚙️ Customisable Settings – Adjust detection range & deterrent intensity.
💻 Easy Monitoring – Web dashboard for instant control & data visualisation.



EFFICIENT LIVE VIDEO FEED

The image shows a dual-screen setup on a Raspberry Pi 4B. The left screen displays a terminal window with system statistics and a process list. The right screen shows a web browser with a 'Live Video Feed' of a bird, with a bounding box and the text 'Birds Approaching: 1'.

Terminal Output:

```
0[|||||] 23.7% Tasks: 94, 248 thr, 139 kthr; 1 runnin
1[|||||] 13.7% load average: 0.53 0.82 0.90
2[|||||] 48.2% Uptime: 00:40:00
3[|||||] 45.8%
Mem[|||||] 1.32G/3.70G
Swp[|||||] 0K/512M
```

PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command
7661	myrios	20	0	606M	56976	43920	S	51.1	1.5	0:00.77	libcamera-sti
810	vnc	20	0	836M	118M	92048	S	13.9	3.1	3:11.65	wayvnc --rend
7199	myrios	20	0	1973M	624M	281M	S	11.9	16.5	0:51.95	/home/myrios/
7224	myrios	20	0	1973M	624M	281M	S	7.3	16.5	0:19.83	/home/myrios/
1109	vnc	20	0	836M	118M	92048	S	5.3	3.1	0:25.73	wayvnc --rend
7662	myrios	20	0	606M	56976	43920	S	4.6	1.5	0:00.07	libcamera-sti
2247	myrios	20	0	8244	3968	2816	R	4.0	0.1	1:25.42	htop
7663	myrios	20	0	606M	56976	43920	S	4.0	1.5	0:00.06	libcamera-sti
7664	myrios	20	0	606M	56976	43920	S	4.0	1.5	0:00.06	libcamera-sti
7667	myrios	20	0	606M	56976	43920	R	3.3	1.5	0:00.05	libcamera-sti
1960	myrios	20	0	531M	40704	30336	S	2.7	1.0	0:40.90	lxtterminal
4707	myrios	20	0	33.4G	142M	96048	S	2.7	3.8	1:21.43	/usr/lib/chro
931	myrios	20	0	556M	102M	80000	S	2.0	2.7	1:15.69	/usr/bin/labw

Web Browser Output:

```
[MQTT] Arduino detected bird presence: False
[0:40:00.017157204] [7661] INFO Camera camera_manager.cpp:327 libcamera v0.4.0+
53-20156679
[0:40:00.063681085] [7667] WARN RPISdn sdn.cpp:40 Using legacy SDN tuning - ple
ase consider moving SDN inside rpi.denoise
[0:40:00.066237516] [7667] INFO RPI vc4.cpp:447 Registered camera /base/soc/12c
0mux/12c@1/ov5647@36 to Unicam device /dev/media4 and ISP device /dev/media6
[0:40:00.066361879] [7667] INFO RPI pipeline_base.cpp:1121 Using configuration
file '/usr/share/libcamera/pipeline/rpi/vc4/rpi_apps.yaml'
Mode selection for 1296:972:12:P
S08RG10_CSI2P,640x480/0 - Score: 3296
S08RG10_CSI2P,1296x972/0 - Score: 1800
S08RG10_CSI2P,1920x1080/0 - Score: 1349.67
S08RG10_CSI2P,2592x1944/0 - Score: 1567
Stream configuration adjusted
[0:40:00.072932454] [7661] INFO Camera camera.cpp:1262 configuring streams: (0)
1296x972-YUV420 (1) 1296x972-S08RG10_CSI2P
[0:40:00.073345275] [7667] INFO RPI vc4.cpp:622 Sensor: /base/soc/12c0mux/12c@1
/ov5647@36 - Selected sensor format: 1296x972-S08RG10_1X10 - Selected unicam for
mat: 1296x972-pdAA
```

Low CPU usage even when YOLO, MQTT, Python Flask & MySQL are running concurrently on the Raspberry Pi 4B



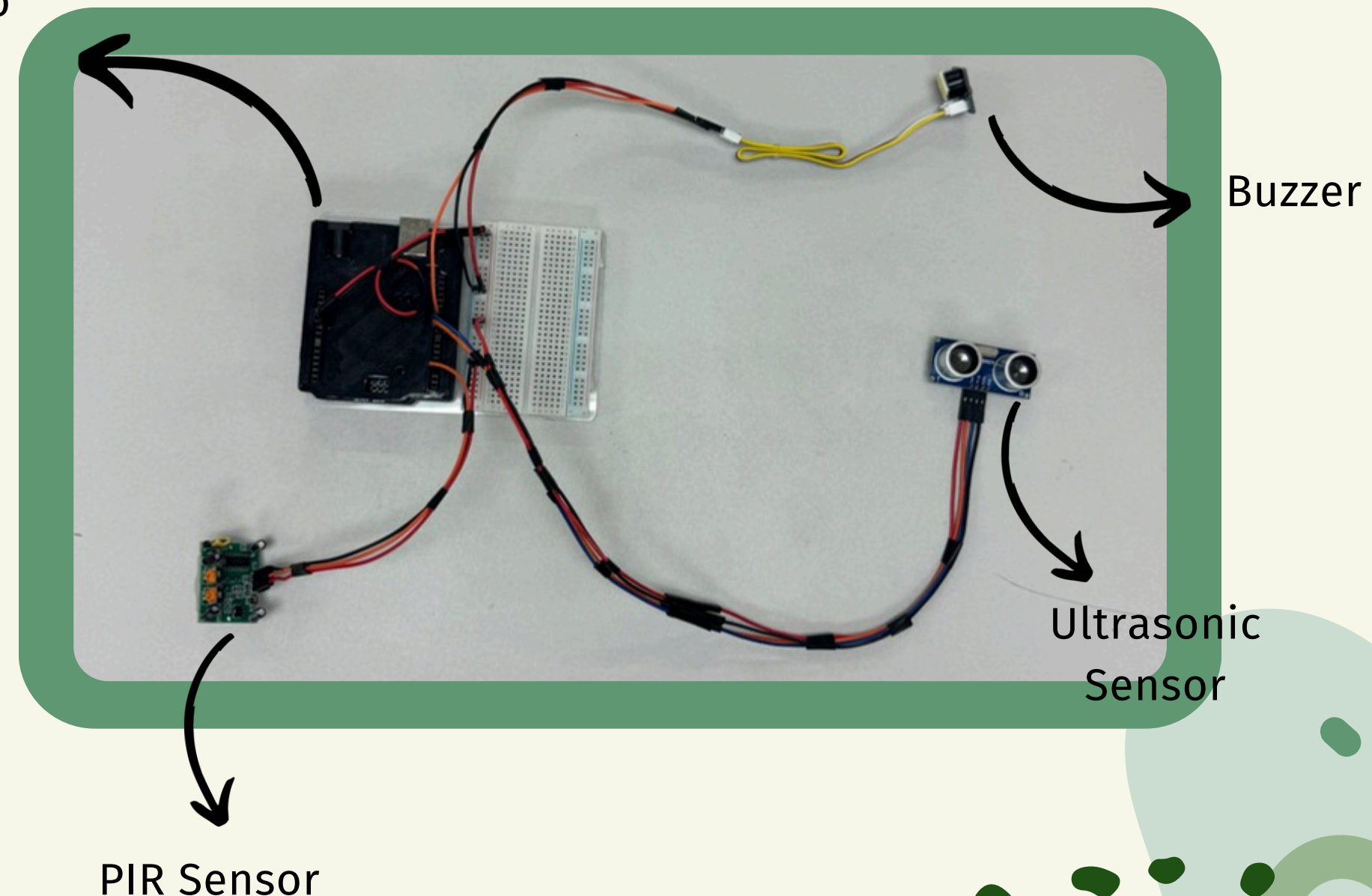
ARDUINO - DETECTION & DETERRENCE MODULE

🎯 Detects & deters birds using ultrasonic & PIR sensors

⚙️ Key Features:

- 📡 Senses movement & communicates via MQTT
- 🔊 Activates buzzer for deterrence
- 🛠️ Customisable: Adjustable range, sound frequency, on/off control

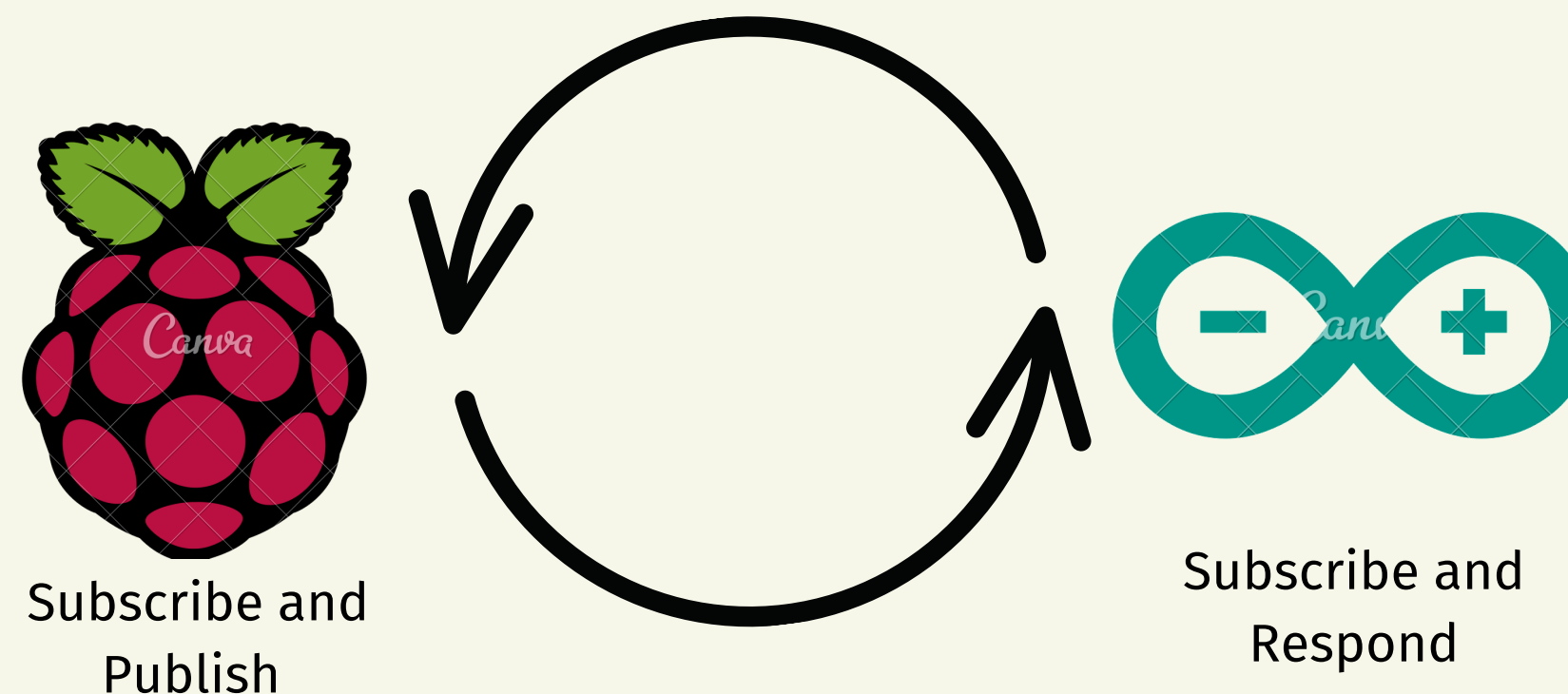
Arduino Uno
Wifi Rev2



MQTT COMMUNICATION IN AVI-DEFENDER

How MQTT Enables Smart Communication

- Raspberry Pi as MQTT Broker & Publisher
 - Controls all message flow and publishes detection & deterrent commands.
- Arduino as Subscriber & Responder
 - Listens for detection data and activates deterrents when necessary.





HOW AVI-DEFENDER COMMUNICATES USING MQTT

🔄 Process Overview: Pi detects bird → Publishes data → Updates settings → Arduino activates buzzer if conditions match → Logs status → Turns off when bird leaves.

Topic	Published By	Subscribed By	Message Type
<code>bird/detection/presence</code>	Raspberry Pi	Arduino	"true" or "false" (bird detected or not)
<code>bird/detection/range</code>	Raspberry Pi	Arduino	Integer (e.g., 50 for 50 cm detection)
<code>buzzer/pitch</code>	Raspberry Pi	Arduino	Integer (e.g., 1500 Hz buzzer frequency)
<code>bird/detection/deterrent</code>	Raspberry Pi	Arduino	"true" or "false" (enable/disable deterrent)
<code>buzzer/activated</code>	Arduino	Raspberry Pi	"true" if buzzer is on, "false" if off

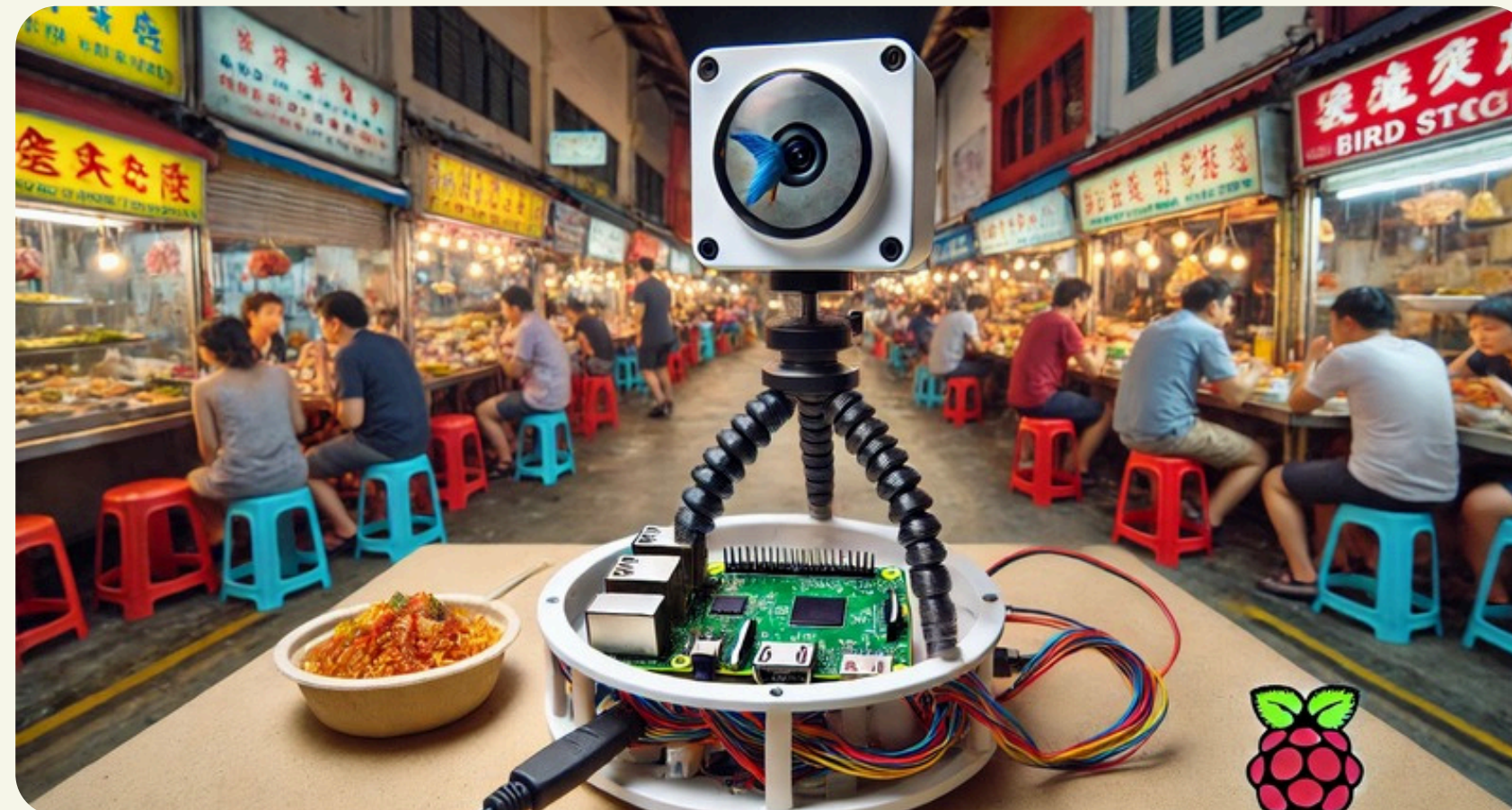
 Table: MQTT Topics & Message Flow



OUR FUTURE IMPROVEMENTS

1. LIVE VIDEO DETECTION - THE NEXT STEP

-  Upgrading camera for continuous real-time monitoring
-  YOLOv11Nano optimised for high-FPS video input
-  Rotating Pi setup enables smooth video feed





2. USING DATA FOR SMART DECISION-MAKING

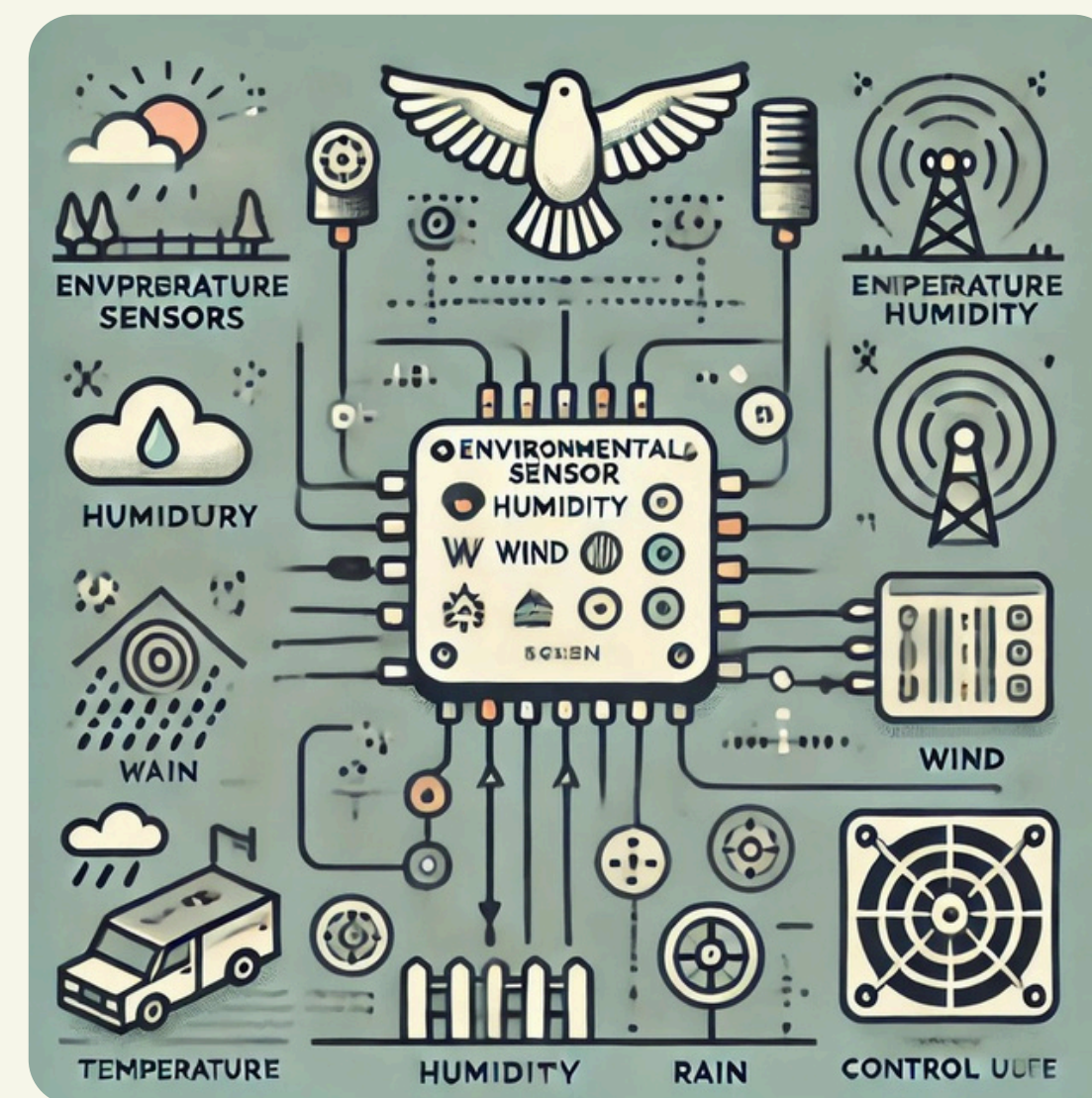
-  Analyze historical bird movement patterns
-  Optimize deterrence times for higher effectiveness
-  Machine learning models for predictive analysis





3. ADDING NOISE & WEATHER SENSORS

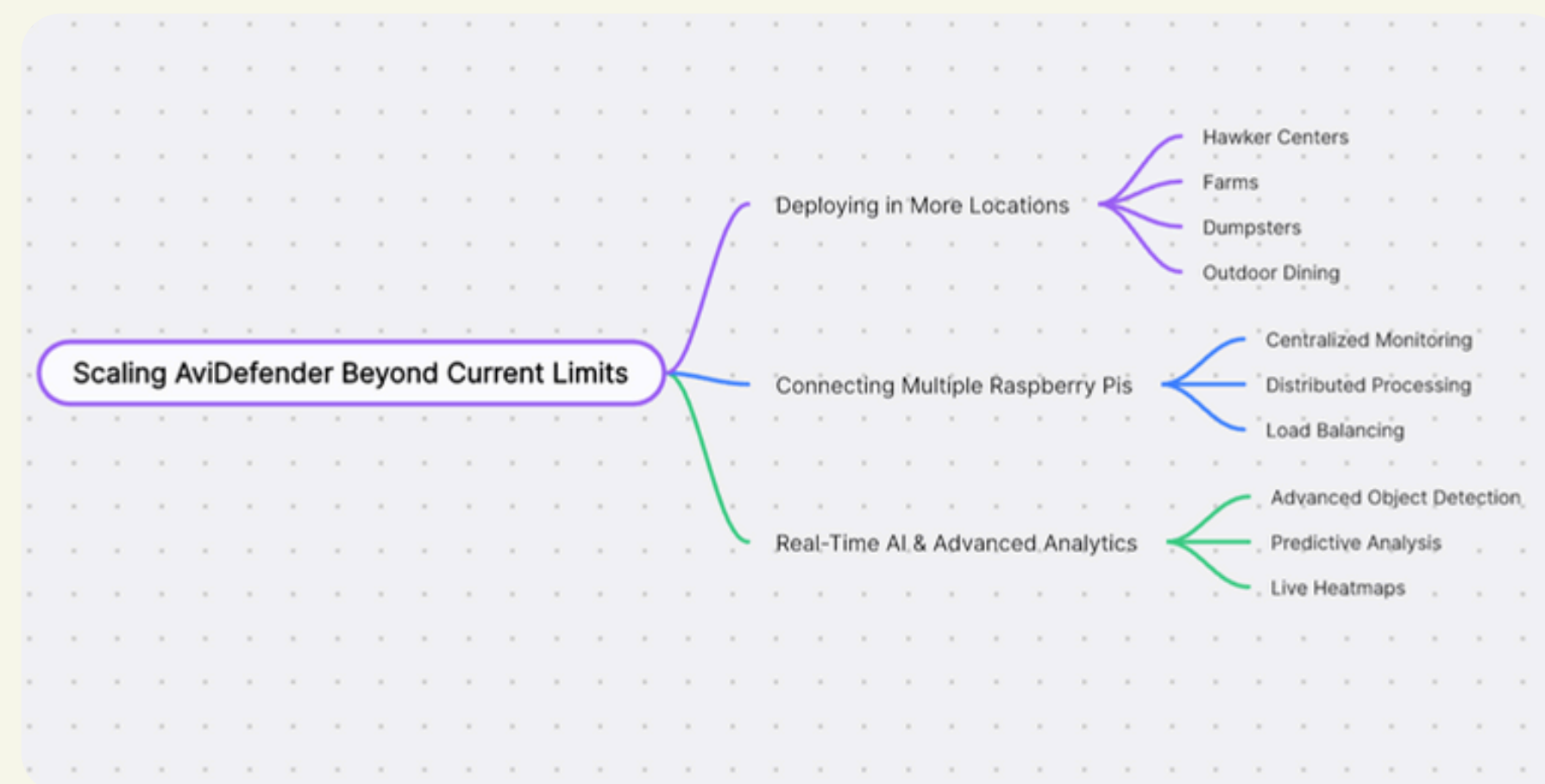
- ☒ Dynamically adjusts deterrence based on weather conditions (whether birds come more or less on a sunny/rainy day)
- ☒ Noise levels influence bird detection accuracy
- ☒ Enhances adaptability across locations





4. SCALING AVI-DEFENDER BEYOND CURRENT LIMITS

-  Deploying in more locations
-  Connecting multiple Raspberry Pis for larger monitoring networks
-  Real-time AI & advanced analytics





DEMONSTRATION



WHY AVI-DEFENDER?

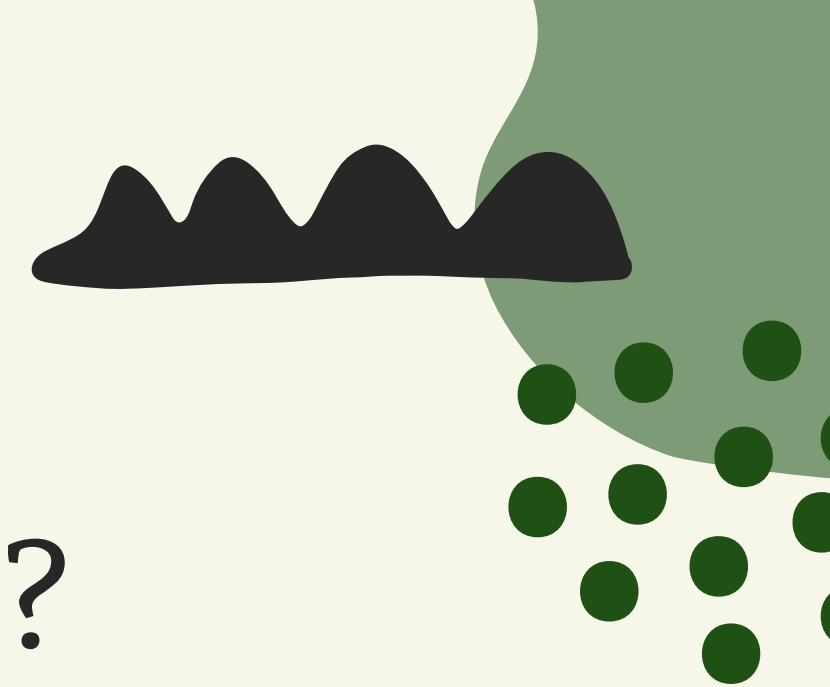
-  AI-driven smart deterrence
-  Scalable & cost-effective
-  Future-proof with AI & video analytics

Before



After





DID WE MEET OUR GOALS/IMPACTS?

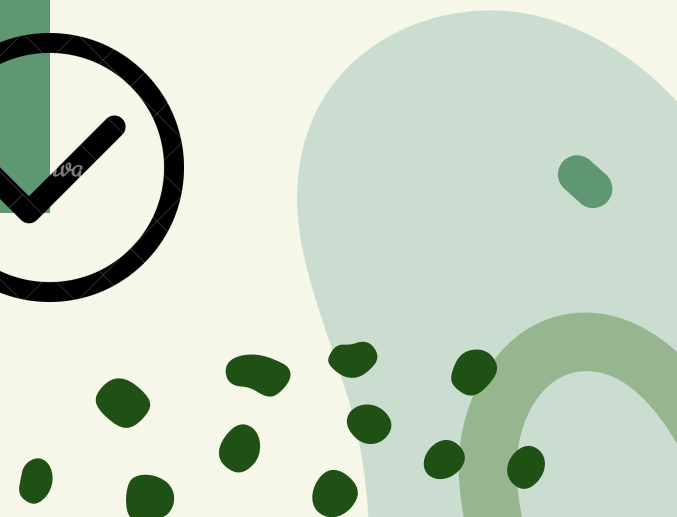
Hygienic Spaces → Prevents contamination with real-time bird deterrence.



Crop Protection → Detects and scares birds before damage occurs.



Efficient Operations → Reduces manual intervention and cleanup.



CONCLUSION

-  AI & sensors detect & deter birds.
-  Prevents contamination, ensuring hygiene.
-  Real-time alerts & data logging for analysis.
-  Raspberry Pi & Arduino communicate via MQTT.
-  Customizable deterrents (range, sound frequency).
-  Future-ready: Sensors, analytics, scalability.



Q&A



THANK
YOU